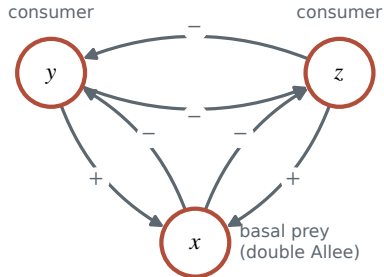
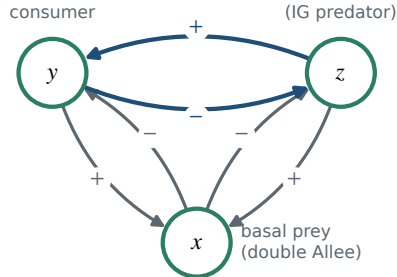


(a) two competing consumers: no-go
competition $y \leftrightarrow z$ ($-c_{12}, -c_{21}$)



$a_{12}a_{23}a_{31} = (-q_1)(-c_{12})(e_2q_2) > 0$
 $a_{13}a_{32}a_{21} = (-q_2)(-c_{21})(e_1q_1) > 0$
 $\Rightarrow L_3 > 0, \kappa = \sum \beta - 2 - L_3 < T_1(\beta)$
 cannot reach $\kappa > T_1$ on the strict-P stratum

(b) intraguild predation: signed 3-cycles
intraguild predation: z eats y ($-h, +e_3h$)
omnivore (IG predator)



$a_{12}a_{23}a_{31} = (-q_1)(-h)(e_2q_2) = +e_2hq_1q_2$
 $a_{13}a_{32}a_{21} = (-q_2)(e_3h)(e_1q_1) = -e_1e_3hq_1q_2$
 $\Rightarrow L_3 = hq_1q_2(e_2 - e_1e_3)/(sc_1c_2) < 0$ if $e_1e_3 > e_2$
 signed 3-cycles lift κ above T_1 : open band reachable